

CHTOURA, Lebanon

WMO: 401010

Lat: 33.817N

Lon: 35.850E

Elev: 920

StdP: 90.75

Time Zone: 2.00 (LBN)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.7	0.7	-9.2	1.9	11.6	-6.6	2.4	11.0	8.3	5.5	7.3	7.4	0.7	50	0.402

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.2	37.2	18.8	35.5	18.6	34.1	18.3	21.2	31.9	20.5	31.2	19.8	30.5	3.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	14.7	24.5	17.1	13.7	25.1	16.0	12.8	25.4	66.2	31.8	63.4	31.1	60.9	30.5	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.3	6.4	5.7	-3.0	39.1	1.6	1.8	-4.2	40.4	-5.1	41.5	-6.0	42.5	-7.2	43.8
			-3.7	22.8	1.5	0.7	-4.8	23.3	-5.7	23.7	-6.5	24.1	-7.7	24.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.1	7.2	8.6	11.5	15.1	19.7	23.6	26.1	26.4	23.6	19.7	13.9	9.0
	DBStd	7.40	3.13	3.47	3.78	3.66	3.32	2.67	2.24	2.18	2.34	3.19	2.91	3.01
	HDD10.0	244	96	62	27	3	0	0	0	0	0	0	3	53
	HDD18.3	1409	345	273	214	107	24	1	0	0	0	22	134	289
	CDD10.0	2830	9	23	72	158	301	409	499	507	407	301	121	22
	CDD18.3	953	0	0	2	12	67	159	241	249	157	64	3	0
	CDH23.3	11148	0	1	18	143	782	1921	3038	3044	1578	593	31	0
	CDH26.7	5376	0	0	1	20	271	889	1637	1672	697	187	2	0
Wind	WSAvg	2.0	1.7	2.1	2.2	2.3	2.2	2.4	2.3	2.1	2.0	1.6	1.3	1.5
Precipitation	PrecAvg	630	139	110	94	42	10	1	0	0	1	33	69	131
	PrecMax	1174	403	244	226	195	40	6	0	1	14	91	221	373
	PrecMin	354	43	28	22	0	0	0	0	0	0	0	0	12
	PrecStd	173	72	55	54	41	12	1	0	3	25	50	79	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.8	22.1	25.4	29.0	34.2	37.2	39.9	39.2	36.0	32.8	26.4	21.7
		MCWB	9.6	11.6	12.6	14.2	16.7	17.5	19.0	19.7	17.8	15.8	13.7	10.4
	2%	DB	16.1	19.0	22.7	26.8	31.8	34.8	37.3	37.2	34.0	30.7	23.9	18.1
		MCWB	8.9	9.9	11.5	13.6	15.9	17.4	18.6	19.3	17.3	15.5	12.4	10.0
	5%	DB	14.1	16.6	20.6	25.1	29.7	33.0	35.3	35.6	32.4	28.8	21.8	15.7
		MCWB	8.0	9.0	10.7	13.3	15.4	17.4	18.6	19.1	17.3	15.2	12.0	9.1
	10%	DB	12.1	14.4	18.5	23.2	27.8	31.5	33.7	34.0	30.7	26.7	19.7	13.7
		MCWB	7.4	8.1	9.9	12.5	14.9	17.1	18.6	19.0	17.1	14.9	11.4	8.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.6	12.1	14.1	15.7	17.9	20.5	22.4	22.5	20.6	18.1	15.1	12.5
		MCDB	16.3	20.5	22.9	24.6	29.8	31.9	33.6	33.6	29.6	27.1	23.0	16.8
	2%	WB	9.6	10.6	12.3	14.6	17.0	19.3	21.1	21.5	19.5	17.0	14.0	11.1
		MCDB	14.6	17.4	20.3	24.5	28.4	31.1	32.5	32.2	29.0	26.7	20.6	15.7
	5%	WB	8.7	9.6	11.4	13.8	16.2	18.5	20.3	20.8	18.7	16.2	13.0	10.1
		MCDB	12.8	15.3	18.9	23.5	27.4	30.1	31.5	31.2	28.6	25.3	19.2	13.8
	10%	WB	7.9	8.5	10.4	13.0	15.5	17.7	19.6	20.1	18.0	15.6	12.2	9.3
		MCDB	11.5	13.3	17.3	22.1	26.2	29.2	30.7	30.6	28.0	24.1	17.8	12.5
Mean Daily Temperature Range	5% DB	MDBR	8.3	8.7	10.1	11.7	13.6	14.5	14.7	15.2	14.4	13.1	11.1	8.7
		MCDBR	12.1	12.9	13.8	14.8	16.0	16.4	17.0	17.1	16.3	15.8	13.7	12.6
		MCWBR	6.2	6.0	6.0	6.4	6.5	6.6	6.9	6.4	6.2	5.9	5.7	6.3
	5% WB	MCDBR	10.2	11.3	12.5	13.7	14.7	15.0	15.3	15.9	14.7	13.7	11.5	9.8
		MCWBR	5.7	5.7	6.1	6.7	6.7	7.1	7.0	6.7	6.1	5.5	5.4	5.4
Clear-Sky Solar Irradiance	taub	0.356	0.393	0.445	0.500	0.492	0.414	0.422	0.433	0.418	0.435	0.381	0.362	
	taud	2.255	2.128	1.974	1.849	1.908	2.168	2.151	2.116	2.146	2.071	2.209	2.240	
	Ebn at Noon	837	841	824	797	807	871	861	846	841	783	799	802	
	Edh at Noon	108	136	172	205	195	150	152	155	143	142	112	104	
All-Sky Solar Radiation	RadAvg	2.59	3.42	4.90	6.14	7.36	8.21	8.03	7.28	6.13	4.49	3.22	2.44	
	RadStd	0.22	0.34	0.30	0.38	0.35	0.14	0.11	0.15	0.17	0.23	0.21	0.22	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (13 neighbors)	+0.30	N/A	-1.33	N/A	N/A	N/A	N/A	N/A	+173	+110

Nomenclature: See separate page